

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – PSEUDOCODE WRITING TASK

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO5 – Naturalization (BL4, BL6)	Assessment:	Assessment Tool 1 – Pseudocode Writing Task
Weightage:	35% of CIA	Total Marks:	100
CO5 Statement:	Construct MST using shortest path algorithms and traverse the graph to model and analyse realworld problem.		
Student Name:	SRIVATSAN L	Reg. No.:	192521480
Section / Batch:	2nd	Date:	28/08/2026

Code of Conduct

I SRIVATSAN L (192521480) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: SRIVATSAN L

Instructions

- This test contains 5 Pseudocode Writing Task questions. Total: 100 Marks.
- When a student answer using pseudocode writing task should evaluate based on the ability to design clear, logical, and efficient pseudocode solutions for data structure problems.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20

Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
GRAND TOTAL:			100 Marks

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and clear understanding; handles edge cases effectively	Good understanding; minor gaps in edge case handling	Partial understanding; misses some requirements	Misinterprets problem; major gaps
Code Correctness	30	Fully correct; passes all test cases	Mostly correct; minor errors	Partially correct; several test cases fail	Incorrect or nonfunctional code
Code Quality & Readability	20	Clean, modular, wellcommented, follows standards	Readable with some structure	Limited readability; poor structure	Unorganized and hard to read
Complexity Analysis	20	Accurate time & space complexity with justification	Correct but limited explanation	Incomplete or partially correct	Missing or incorrect
Testing & Validation	10	Thorough testing including edge cases	Adequate testing with few edge cases	Minimal testing	No testing evidence
Faculty In-charge		Course Coordinator		Program Director	
Signature: _____		Signature: _____		Signature: _____	
Date: _____		Date: _____		Date: _____	

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QUESTIONS
Q1
Minimum Spanning Tree -
Kruskal's Algorithm
20 marks

Q2
Graph - Topological Sort
20 marks

Q3
Shortest Path Algorithm - MST
& Dijkstra's Algorithm
20 marks

Q4
Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q1 Minimum Spanning Tree - Kruskal's Algorithm

20 marks

Write pseudocode for constructing an MST using Kruskal's Algorithm. A city planning department needs to connect multiple zones with underground fiber-optic cables. Each zone is represented as a vertex, and possible cable connections between zones are represented as edges with installation costs (weights).

Constraints:

- All zones must be connected
- Total installation cost must be minimized
- No redundant connections (no cycles) allowed
- The network must remain fully connected

Your Answer

```
MAKE_SET(x)
parent[x] = x
rank[x] = 0
```

```
FIND_SET(x)
if parent[x] != x:
```

Save Answer

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QUESTIONS
Q1
Minimum Spanning Tree -
Kruskal's Algorithm
20 marks

Q2
Graph - Topological Sort
20 marks

Q3
Shortest Path Algorithm - MST
& Dijkstra's Algorithm
20 marks

Q4
Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q2 Graph - Topological Sort

20 marks

A DAG represents task dependencies. Write pseudocode for Topological Sort using DFS with cycle detection.

Constraints:

- Detect cycle before sorting
- Use DFS approach

Your Answer

```
FUNCTION topologicalSort(graph):
// Initialize all states to UNVISITED (0)
state = Map()
FOR EACH vertex v IN graph.vertices:
state[v] = 0
```

Save Answer

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QUESTIONS

Q1 ✓

Minimum Spanning Tree -
Kruskal's Algorithm
20 marks

Q2 ✓

Graph - Topological Sort
20 marks

Q3 ✓

Shortest Path Algorithm - MST
& Dijkstra's Algorithm
20 marks

Q4 ✓

Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5 ✓

Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

Q3 Shortest Path Algorithm - MST & Dijkstra's Algorithm

20 marks

Write pseudocode integrating MST + Dijkstra for Smart transportation system.

Constraints:

- a. Use MST for infrastructure
- b. Use shortest path for routing

Your Answer

```
// Step 1: Define Graph Structure
Graph G = (V, E) // V = intersections/hubs, E = all possible road segments with weights (cost/distance)
```

Function SmartTransportationSystem(Graph G, Node source, Node destination):

```
// Constraint A: Use MST for Infrastructure Backbone
```

Save Answer

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QUESTIONS

Q1

Minimum Spanning Tree -
Kruskal's Algorithm
20 marks

Q2

Graph - Topological Sort
20 marks

Q3

Shortest Path Algorithm - MST
& Dijkstra's Algorithm
20 marks

Q4

Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5

Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q4 Shortest Path Algorithm - Dijkstra's Algorithm

20 marks

Shortest Path with "Toll-Free" Voucher: You have a voucher that allows you to traverse one edge for free (weight = 0). Write pseudocode to find the shortest path from S to D utilizing this voucher optimally.

Your Answer

```
Function ShortestPathWithVoucher(Graph, S, D):  
    // Graph is represented as an adjacency list where Graph[u] contains pairs (v, weight)  
  
    Initialize a 2D array dist[][] with infinity (∞) for all nodes and states  
    Initialize a priority queue PQ storing elements of the form (current_distance, current_node, voucher_used)
```

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QUESTIONS
Q1 ✓
Minimum Spanning Tree -
Kruskal's Algorithm
20 marks

Q2 ✓
Graph - Topological Sort
20 marks

Q3 ✓
Shortest Path Algorithm - MST
& Dijkstra's Algorithm
20 marks

Q4 ✓
Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5 ✓
Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q5 Minimum Spanning Tree - Prim's or Kruskal's Algorithm

20 marks

Second Best Spanning Tree: Design pseudocode to find the "Second Best" MST. (20 Marks)
Hint: Find the MST first, then for every edge in the MST, try removing it and finding the next best edge to replace it.

Your Answer

```
Algorithm FindSecondBestMST(Graph G):
Input: A connected, weighted, undirected graph G = (V, E)
Output: The weight and the set of edges of the Second Best MST

// Step 1: Find the initial Minimum Spanning Tree
MST_edges = KruskalOrPrim(G)
```

[Save Answer](#)